

Digital Maturity Model

48 capabilities across 8 layers, each rated 0 to 4. Rate every capability against routine use by its intended users — not against sanction, procurement or deployment alone. Tick one box per capability, then carry the layer subtotals to the score sheet at the end.

STATE / UT

FILLED BY · DESIGNATION

DATE

THE RATING SCALE

0	Does not exist	No digital system for this capability. Handled on paper, in spreadsheets, or not at all.
1	Under development	A system has been sanctioned, procured or is in development. Nothing is in productive use yet.
2	Pilot complete	Live and working with real data in a limited setting — one or a few districts, or a small user group — for at least one full reporting cycle. Not yet rolled out further.
3	Functional at limited scale	In routine production use across part of the state by the intended users. Gaps remain — some districts still manual, or some workflow steps still offline.
4	Fully functional at state scale	In routine use across effectively all districts by effectively all intended users. The data is authoritative, no parallel manual register is maintained, and the system is owned and monitored.

Levels 3 and 4 depend on routine **use** by the intended users, not on rollout alone — a system live statewide but used in three districts is a pilot (2). If a parallel manual register is still the record of truth, the capability is not at 4.

1 Feedback & Satisfaction

Whether the State has the digital capability to systematically collect, analyze, and act upon citizen feedback regarding rural drinking water services.

INCLUDES Feedback capture · Sentiment analysis · Closure of loop

0	1	2	3	4
☐	☐	☐	☐	☐

2 Consumer Registry & Profile

Whether the State has a digital registry of FHTC / consumers with a unique identifier, and whether this database is authoritative, complete, up-to-date, interoperable and actively used.

INCLUDES Unique consumer ID · Household mapping · Interoperability · Active use

0	1	2	3	4
☐	☐	☐	☐	☐

3 Connection Lifecycle Management

Whether the State effectively manages the entire lifecycle of a Functional Household Tap Connection (FHTC): from application through installation, modification, transfer, suspension and disconnection.

INCLUDES Application · Installation · Transfer · Suspension · Disconnection

0	1	2	3	4
☐	☐	☐	☐	☐

4 Digital Billing & Payments

Whether the state has the capability to digitally view bills, receive payment reminders, make online payments through multiple channels, access payment history and receive digital receipts.

INCLUDES Bill view · Reminders · Online payment · Payment history · Digital receipts

0	1	2	3	4
☐	☐	☐	☐	☐

5 Notifications & Alerts

Whether the system has the capability to notify citizens before they need to ask — planned supply interruptions, water quality advisories, bill due reminders, complaint status updates or restoration notices.

INCLUDES Supply interruptions · Quality advisories · Bill reminders · Complaint status

0	1	2	3	4
☐	☐	☐	☐	☐

6 Citizen Engagement & Participation

Whether the State provides digital mechanisms that enable citizens and communities to actively participate in governance, monitoring, planning and decision-making.

INCLUDES Participation · Community monitoring · Planning inputs

0	1	2	3	4
☐	☐	☐	☐	☐

Layer 1 subtotal — Citizens sum of six ratings

	/ 24
----------------------	------

2 Frontline Workers

Digital tools used by field staff for monitoring and reporting

7 Water Quantity & Regularity Monitoring

Whether the system has the capability to enable frontline workers to accurately measure, digitally report and validate daily water supply data in a timely manner using standardized processes and digital tools.

INCLUDES Daily reporting · Validation · Integration · Governance

0	1	2	3	4
☐	☐	☐	☐	☐

8 Water Quality Monitoring & FTK

Whether the system has the capability to enable frontline workers to conduct Field Test Kit (FTK) tests, digitally report water quality observations, identify potential quality issues and support routine village-level surveillance.

INCLUDES FTK testing · Digital reporting · Issue identification

0	1	2	3	4
☐	☐	☐	☐	☐

9 Asset Maintenance & Financial Management

Whether the system has the capability to enable frontline workers to digitally record asset maintenance, O&M expenditure, collections and village-level financial management.

INCLUDES Maintenance logs · O&M expenditure · Collections

0	1	2	3	4
☐	☐	☐	☐	☐

10 Community Engagement

Whether the system has the capability to enable frontline workers to digitally plan, conduct, record and monitor community mobilization and IEC activities that promote safe water practices and sustainable operation.

INCLUDES IEC planning · Activity records · Monitoring

0	1	2	3	4
☐	☐	☐	☐	☐

11 Field Data Management

Whether the system has the capability to enable frontline workers to digitally collect, validate, update and report operational field data, inspections, surveys, asset information and compliance records.

INCLUDES Field collection · Inspections · Surveys · Compliance records

0	1	2	3	4
☐	☐	☐	☐	☐

12 Digital Work Assistance

Whether the State has the capability to digitally empower frontline workers through mobile applications, role-based dashboards, contextual digital assistants, training, knowledge resources and alerts.

INCLUDES Mobile apps · Role dashboards · Training · Alerts

0	1	2	3	4
☐	☐	☐	☐	☐

Layer 2 subtotal — Frontline Workers sum of six ratings

	/ 24
----------------------	------

13 Project Delivery & Quality Assurance

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally plan, execute, monitor and validate rural water infrastructure projects, ensuring timely completion, adherence to quality standards and compliance with contractual and technical requirements.

INCLUDES Planning · Execution · Quality checks · Compliance

14 Asset Lifecycle Management

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally manage water supply assets throughout their lifecycle, including commissioning, tagging, handover, maintenance history, condition monitoring and lifecycle updates.

INCLUDES Commissioning · Tagging · Handover · Maintenance history · Condition monitoring

15 Water Service Delivery

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally operate and manage rural water supply schemes by monitoring water production, bulk water flow, supply schedules, service continuity and operational performance.

INCLUDES Production · Bulk flow · Supply schedules · Service continuity

16 Water Quality Management

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally manage water quality surveillance through laboratory testing, compliance monitoring, reporting and timely corrective actions to ensure safe and compliant drinking water.

INCLUDES Lab testing · Compliance monitoring · Reporting · Corrective action

17 Operations & Resource Management

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally plan, allocate and monitor operational resources, including workforce, equipment, materials, consumables and finances, to support efficient O&M of water supply schemes.

INCLUDES Workforce · Equipment · Materials · Finances

18 Contract Performance & Compliance

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally monitor contractual obligations, service levels, inspections, audits, performance indicators and statutory compliance to ensure effective and accountable service delivery.

INCLUDES Obligations · Service levels · Inspections · Audits · Statutory compliance

Layer 3 subtotal — Agencies sum of six ratings

	/ 24
----------------------	------

19 Infrastructure Planning & Asset Governance

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally plan, design, govern and manage rural water infrastructure throughout its lifecycle, including DPR preparation, asset inventory, commissioning and lifecycle management.

INCLUDES Asset Management · Inventory · BOQ · DPR · Asset Registry

20 Vendor & Financial Management

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally manage procurement, contracts, work orders, invoices, payments and financial transactions, ensuring financial accountability and integration with government financial systems.

INCLUDES Vendor Management · Contracts · Work Orders · Invoices · Payments · IFMS

21 Water Resource & Sustainability Management

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally assess, monitor and manage water sources, hydrogeology and source sustainability to ensure long-term availability and resilience.

INCLUDES Source Sustainability · Source Geology · Water Audits

22 Water Service Intelligence

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally monitor, analyse and optimise rural water service delivery using operational data, dashboards, analytics, predictive insights and decision-support tools.

INCLUDES BI · MIS · Dashboards · Predictive Maintenance · Scheme Risk Forecasting

23 Water Quality Governance

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally monitor, analyse and govern drinking water quality through laboratory management, surveillance, compliance monitoring and timely corrective actions.

INCLUDES LIMS · Water Quality surveillance

24 Engineering & Network Intelligence

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to digitally model, monitor and optimise water supply infrastructure using engineering tools, spatial intelligence, network modelling and operational telemetry.

INCLUDES GIS · Geo-tagging · Hydraulic Modeling · IoT · SCADA

Layer 4 subtotal — Department sum of six ratings

	/ 24
----------------------	------

25 Interdepartmental Coordination & Collaboration

Whether the State has the capability to digitally coordinate, collaborate and exchange information across departments to enable seamless execution of programmes and coordinated governance of rural drinking water services.

INCLUDES Data exchange · Workflow · Coordination

0	1	2	3	4
☐	☐	☐	☐	☐

26 Integrated Planning & Programme Governance

Whether the State has the capability to digitally support joint planning, approvals, programme monitoring and governance across multiple departments.

INCLUDES Joint planning · Approvals · Programme governance

0	1	2	3	4
☐	☐	☐	☐	☐

27 Cross-Sector Policy & Regulatory Alignment

Whether the State has the capability to digitally support alignment of policies, regulations, standards and institutional responsibilities across sectors.

INCLUDES Policy alignment · Regulatory coordination

0	1	2	3	4
☐	☐	☐	☐	☐

28 Strategic Intelligence & Decision Support

Whether the State has the capability to provide integrated analytics, dashboards and cross-sector insights that enable evidence-based planning, monitoring and strategic decision-making.

INCLUDES Analytics · Dashboards · Performance monitoring

0	1	2	3	4
☐	☐	☐	☐	☐

29 Public Feedback & Stakeholder Engagement

Whether the State has the capability to digitally consolidate citizen feedback, grievances and stakeholder inputs across departments to improve service delivery, transparency and policy effectiveness.

INCLUDES Grievances · Feedback · Citizen inputs

0	1	2	3	4
☐	☐	☐	☐	☐

30 Emergency Preparedness & Resilience Management

Whether the State has the capability to digitally coordinate preparedness, emergency response, disaster management and recovery across departments to ensure continuity of drinking water services.

INCLUDES Disaster management · Emergency response · Drought · Floods · Contamination events

0	1	2	3	4
☐	☐	☐	☐	☐

Layer 5 subtotal — State Functionaries sum of six ratings

	/ 24
----------------------	------

6 Shared Digital Services

Services shared across all stakeholder groups

31 Stakeholder Identity & Lifecycle Management

Whether the State has the capability to digitally onboard, authenticate, authorize and manage the lifecycle, profiles and access of all stakeholders across the water service ecosystem.

INCLUDES User management · Onboarding · Identity · Roles · Access

0	1	2	3	4
☐	☐	☐	☐	☐

32 Grievance & Service Request Management

Whether the State has the capability to digitally receive, assign, track, escalate and resolve grievances and service requests across stakeholders with timely response, transparency and accountability.

INCLUDES Intake · Assignment · Escalation · Resolution · SLA tracking

0	1	2	3	4
☐	☐	☐	☐	☐

33 Capacity Building & Knowledge Management

Whether the State has the capability to digitally deliver training, certifications, knowledge resources, SOPs and continuous learning opportunities to strengthen stakeholder competencies.

INCLUDES LMS · Training · Certification · Knowledge · Digital learning

0	1	2	3	4
☐	☐	☐	☐	☐

34 Personalized Dashboards & Reports

Whether the State has the capability to provide stakeholders with role-based dashboards, reports, alerts and insights that support operational monitoring, performance management and informed decision-making.

INCLUDES Role-based dashboards · Personalized KPIs · Drill-down · Self-service reporting · Automated distribution

0	1	2	3	4
☐	☐	☐	☐	☐

35 Intelligent Assistance & Decision Support

Whether the State has the capability to provide contextual assistance, conversational interfaces, recommendations and guided workflows that help stakeholders perform tasks more effectively.

INCLUDES Context-aware guidance · Decision support · Recommendations · Conversational assistance

0	1	2	3	4
☐	☐	☐	☐	☐

36 Omnichannel Digital Experience

Whether the State has the capability to enable stakeholders to access digital services through multiple channels such as mobile applications, web portals, WhatsApp, SMS, IVRS and future digital interfaces.

INCLUDES WhatsApp · Mobile · Portal · IVRS · SMS · Email

0	1	2	3	4
☐	☐	☐	☐	☐

Layer 6 subtotal — Shared Digital Services sum of six ratings

	/ 24
----------------------	------

37 Digital Platform & Architecture

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to build and operate scalable, modular, resilient and maintainable digital solutions using modern architectural principles, reusable components and cloud-ready technologies.

INCLUDES Unified platform · Modular architecture · Microservices · Cloud native · DevOps · Containerization

38 Interoperability & Open Integration

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to seamlessly exchange information and integrate with central, state and external systems through open standards, APIs, common data models and interoperable interfaces.

INCLUDES APIs · Centre systems · State systems · Standards · Registries · Eventing

39 Data Management & Governance

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to establish trusted, high-quality and well-governed data through standardized data models, master data management, metadata, data quality and data governance practices.

INCLUDES Master data management · Metadata · Data quality · Data governance · Data catalog · Lineage · Stewardship

40 Digital Identity & Access Management

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to securely manage digital identities, authentication, authorization and access control for users, applications and services while ensuring privacy and accountability.

INCLUDES IAM · SSO · RBAC · MFA

41 Intelligent Automation & Decision Support

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to leverage intelligent technologies to automate business processes, generate insights, support decision-making and improve operational efficiency.

INCLUDES AI · ML · OCR · Prediction · Recommendations · NLP

42 Security, Privacy & Compliance

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to protect digital systems and data through comprehensive security controls, privacy safeguards, risk management and compliance with applicable standards and regulations.

INCLUDES Security · Privacy · Compliance · Audit · Encryption · Monitoring

Layer 7 subtotal — Technology Foundation sum of six ratings

	/ 24
----------------------	------

8 Infrastructure Foundation

The compute and network layer underlying the technology foundation

43 Compute & Hosting Infrastructure

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to provision, manage and optimize reliable compute and hosting infrastructure supporting secure, efficient and continuous operation of digital services.

INCLUDES Cloud · On-premises · Virtualization · Containers · Compute resources

44 Data Infrastructure

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to provide reliable, scalable and high-performance storage, databases and data services that support secure data management, processing and retrieval.

INCLUDES Databases · Object storage · File storage · Caching · Data services

45 Network & Connectivity

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to provide secure, reliable and high-performance network connectivity enabling seamless communication between users, applications, infrastructure and external systems.

INCLUDES Internet · VPN · MPLS · SD-WAN · Connectivity to remote areas

46 Infrastructure Resilience & Continuity

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to ensure uninterrupted operation and rapid recovery of digital services through high availability, backup, disaster recovery and business continuity mechanisms.

INCLUDES HA · Backup · Disaster Recovery · Failover · Business Continuity

47 Infrastructure Monitoring & Operations

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to continuously monitor infrastructure health, performance, availability and operational events, enabling proactive detection, incident response and optimization.

INCLUDES Monitoring · Logging · Alerting · Observability · Incident Management

48 Elasticity & Performance Management

0	1	2	3	4
☐	☐	☐	☐	☐

Whether the State has the capability to dynamically scale infrastructure resources and optimize system performance to meet changing workloads while maintaining reliability and cost efficiency.

INCLUDES Horizontal scaling · Vertical scaling · Auto scaling · Load balancing · Performance tuning

Layer 8 subtotal — Infrastructure Foundation sum of six ratings

	/ 24
----------------------	------

Score sheet

Carry each layer subtotal here, total them, then read the maturity index below.

LAYER	SUBTOTAL	MAX
1 Citizens		24
2 Frontline Workers		24
3 Agencies		24
4 Department		24
5 State Functionaries		24
6 Shared Digital Services		24
7 Technology Foundation		24
8 Infrastructure Foundation		24
Total		192

OVERALL MATURITY INDEX

Total ÷ 192 × 100 = _____ %

- 0–20%
- 21–40%
- 41–60%
- 61–80%
- 81–100%

- Nascent
- Emerging
- Developing
- Mature
- Leading

LAYER MATURITY INDEX

Read each layer subtotal out of 24 directly.

- 0–4 Nascent
- 5–10 Emerging
- 11–15 Developing
- 16–20 Mature
- 21–24 Leading

NOTES · EVIDENCE · SYSTEMS REFERENCED